ATM Transaction Sequence Simulation (Case Study 13)

Problem Statement

Process a list of deposits and withdrawals, update the account balance, and reject withdrawals that exceed the available
balance.

Objectives

Practise lists, loops, conditions, and running totals.

Assumptions

The starting balance is non-negative. Transactions are entered as positive amounts; D means deposit and W means
withdrawal.

Algorithm

1. Read starting balance and number of transactions. 2. For each transaction, read type and amount. 3. Add deposits. 4.
Allow a withdrawal only if the amount is not greater than the balance. 5. Display the final balance.

Flowchart (Text Form)

START → Read balance → Repeat for each transaction → Deposit? add amount : check withdrawal balance →
Update/reject → Print balance → STOP.

Pseudocode

```
E
```

Python Source Code

```
balance = float(input("Starting balance: "))
n = int(input("Number of transactions: "))
for i in range(n):
    kind = input("Transaction D/W: ").upper()
    amount = float(input("Amount: "))
    if amount &lt;= 0:
        print("Invalid amount")
    elif kind == "D":
        balance += amount
        print("Deposit accepted")
    elif kind == "W":
        if amount &lt;= balance:
            balance -= amount
            print("Withdrawal accepted")
        else:
            print("Withdrawal rejected: low balance")
    else:
        print("Invalid transaction type")
print("Final balance:", round(balance, 2))
```

Test Case and Expected Output

Input: balance=1000; transactions D 200, W 500
Expected output: Final balance: 700.0

Conclusion

The program keeps a running balance and prevents withdrawals when funds are insufficient.